

CONTACT INFO

 Niroomand2355@gmail.com  [Github](#)  [Homepage](#)  [Linkedin](#)

RESEARCH INTERESTS

Machine Learning Ensemble Learning HPC Parallel computing GPGPU

EDUCATION

- Jan 2018 - Sep 2022 **Master of Computer Science - Artificial Intelligence**
Institute for Advanced Studies in Basic Sciences (IASBS) , Zanjan, Iran
Thesis: "Parallel Random Forest Implementations Using GPU and CPUs: A Survey"
Supervisors: Dr. Mahmoud Shirazi, Dr. Parvin Razzaghi
Related Courses: Advanced Data Mining, Machine Learning, Parallel Computing
GPA: 72.45/100
- Jul 2013 - Nov 2018 **Bachelor of Software Engineering**
Ayatollah Borujerdi University (ABRU) , Borujerd, Iran
Thesis: "Text Recognition using Augmented Reality"
Supervisor: Dr. Mansour Shahidi
Related Courses: Computer Vision, Advanced programming, Computer Graphics, Image Processing
GPA: 76.4/100

PROJECTS

- Nov 2024 **Tagline Generator** |  [Github](#)
A tagline generator using generative AI tools available in Google Cloud.
[Generative AI](#) [Cloud](#) [Vertex AI Studio](#)

TEACHING EXPERIENCES

- Mar-Oct 2023 **Teaching Assistant, Parallel Computing Course, Institute for Advanced Studies in Basic Sciences (IASBS), A,**
 > Professor: Dr. Mahmoud Shirazi
 > Assisted in teaching the Parallel Computing course with a focus on Open-MPI programming.
 > Source: Parallel and High Performance Computing | Robert Robey and Yuliana Zamora
[Parallel Algorithms](#) [Parallel Programming](#) [Open-MPI](#) [OpenMP](#) [High Performance Computing](#)

WORK EXPERIENCES

- Mar 2023 - Feb 2024 **Research Assistant , Institute for Advanced Studies in Basic Sciences (IASBS)** 
Focused on converting my master's thesis into a comprehensive survey paper and Analyzed GPU-accelerated Random Forest algorithms for performance enhancements.
[Research](#) [Machine Learning](#) [CUDA Programming](#) [GPU Computing](#)

PUBLICATIONS

- Nov 2024, B. Niroomand, S. Pirhadi, M. Shirazi "A Comparative Analysis of Ensemble Learning Methods for Classification Tasks on Benchmark Datasets", submitted to CSICC2025: "29th International Computer Conference - Computer Society of Iran", <https://csicc2025.csi.org.ir/en/>
- Dec 2024, S. Pirhadi, B. Niroomand, E. Ansari, "Impact of dataset size on relation classification models", submitted to CSICC2025: "29th International Computer Conference - Computer Society of Iran", <https://csicc2025.csi.org.ir/en/>
- Sep 2024, B. Niroomand, S. Pirhadi, M. Shirazi, P. Razzaghi, "Parallel Random Forest Implementations Using GPU and CPUs: A Survey", To be submitted to: "International Conference on Parallel Processing", <https://icpp2024.org>

★ HONORS AND AWARDS

- Sep 2018 Ranked among top 10 percent of about 3,000 participants in the Nationwide University Entrance Exam of Master Degree for Computer Science.
- Sep 2022 The IASBS awarded as one of the top advanced research centers in Iran which was also ranked among the IRAN's top universities ranked by QS.

☰ KNOWLEDGE AND SKILLS

- Knowledge** Machine Learning, Deep Learning, Mathematics, Statistics, Calculus, Algebra, Operating Systems, Advanced Databases, Advanced Algorithms, Advanced Data Mining, Parallel Computing
- Programming** C, C++, CUDA, Python, MATLAB, R, Assembly (RISC, x86), Pytorch, Tensorflow, Keras, Sckit-learn, Scipy, Pandas, Numpy, Matplotlib, OpenCV
- DevOps** Docker, Kubernetes, Unix, Bash Script
- Tools** Git, ~~TeX~~TeX, Kaggle
- Soft Skills** Scientific Research, Problem-solving, Presentation skills, Strategic thinking, Critical thinking, Teamwork, Growth mindset

🗣️ LANGUAGES

Persian: Native

English: Advanced (IELTS Exam scheduled for December 2024)

Latin: Basic

Spanish: Basic

Italian: Basic

* ACHIEVEMENTS

- Sep 2024 Introduction to Concurrent Programming with GPUs, The Johns Hopkins University 🔗
- Sep 2024 Introduction to Parallel Programming with CUDA, The Johns Hopkins University 🔗
- Oct 2024 Prompt Design in Vertex AI (Skill Badge), Google Cloud Skill Boost 🔗
- Oct 2024 Various Google Cloud Completion Badges (Mostly Related to AI), Google Cloud Skill Boost 🔗

👤 VOLUNTEERING

- Sep 2013 Board member in Student's Scientific Association of Computer Engineering of ABRU University
- Mar 2019 Local Chair member in The International Conference on Contemporary Issues in Data Science (CIDAS) IASBS
- Jan 2023 Online Attendee of NVIDIA GTC 2023, Participated in online workshops
- Sep 2024 Member of Google Cloud Skills Boost
- Oct 2024 Member of Google Developers Program, Public Profile: <https://g.dev/Babak2355>
- Oct 2024 Member of NVIDIA Developers Program
- Nov 2024 Member of GitHub Developers Program